Magnetic Properties 1

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Week 13 lectures

Magnetic properties

- Magnetic phenomena
- Classification of magnetic materials
- Magnetic behaviour and applications

Optical properties

- Interactions of waves and atoms
- Optical phenomena
- Applications in lasers and optical fibres

Background information:

For these lectures read

Callister and Rethwisch 9e Chapter 21 and Chapter 22

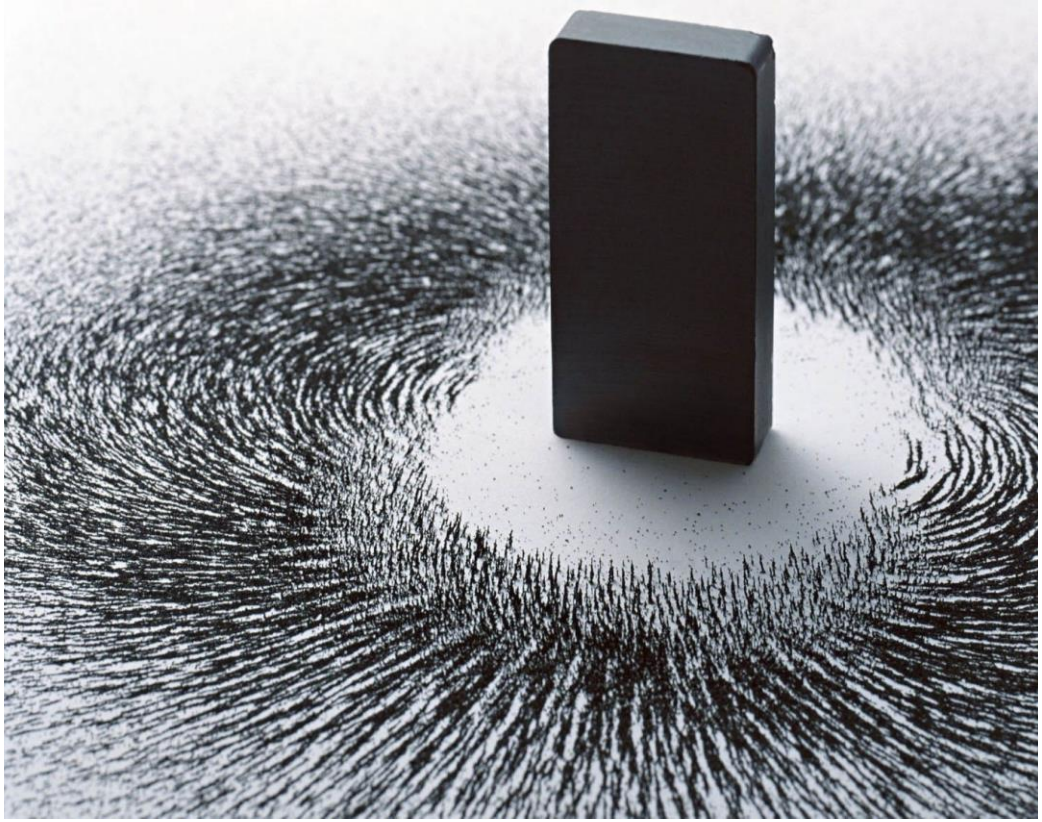
Magnetic behaviour

- **Magnetism**

Attractive and repulsive forces resulting from movement of electronic charge at the atomic scale

Often associated with iron but other materials also have magnetic behaviours

Magnetic forces are vectors

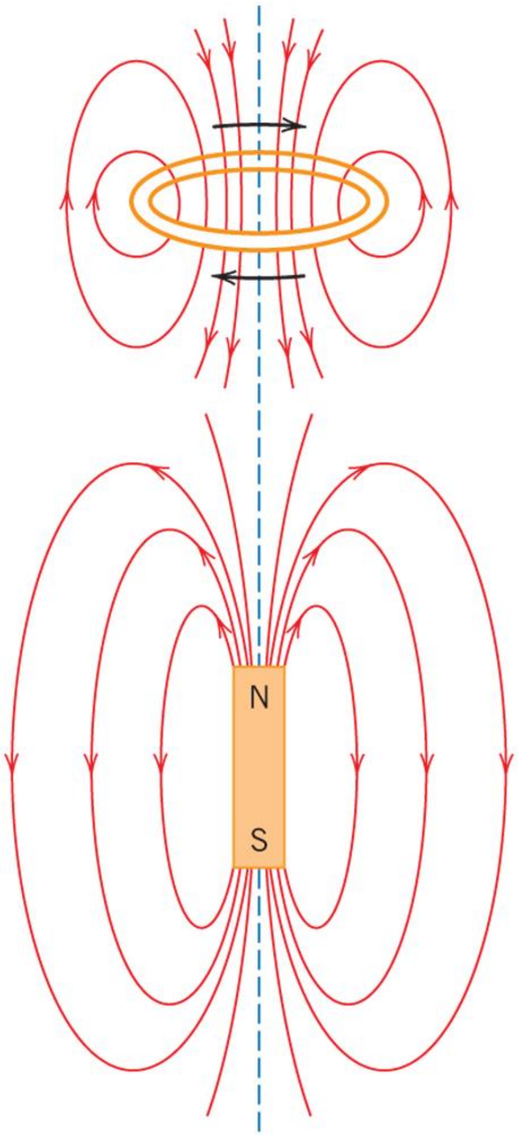


Magnetic behaviour

- **Magnetic dipoles**

- Magnetic forces occur due to movement of electrically charged particles
- By convention, these are represented by lines drawn to indicate the direction of the force (vector)
- Magnetic field lines of force around **(a)** a current loop and **(b)** a bar magnet are shown in the figure

Magnetic field position and direction

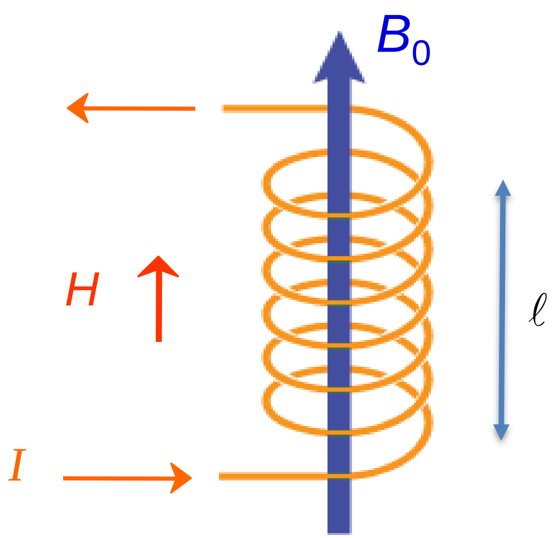


(a)

(b)

Generation of magnetic field

- **In vacuum**
- Created by electric current through a coil



N = total number of turns of wire

ℓ = length (m)

I = current (ampere)

H = **applied magnetic field** (ampere-turns/m)

B_0 = **magnetic flux density** in a vacuum (units of Tesla)

Generation of magnetic field

- **In vacuum**
- Calculation of the **applied magnetic field**, ***H***:

Number of turns

Current

Magnetic field strength

$$H = \frac{N\,I}{\ell}$$

Length of coil

Unit: ampere-turns/meter,
or amperes/meter

- Calculation of the **magnetic flux density** in a vacuum, $\boldsymbol{B_0}$:

$$B_0 = \mu_0 H$$

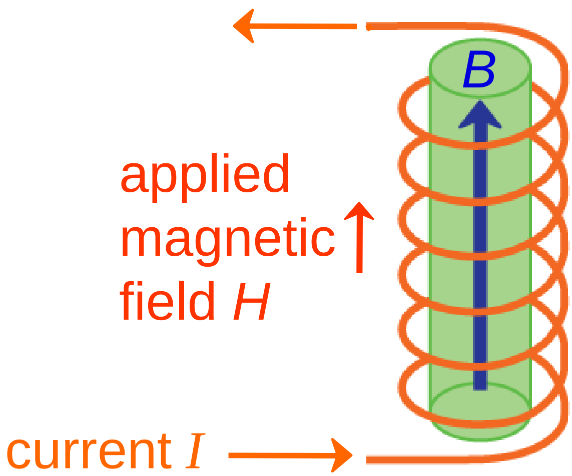
Unit: Tesla

permeability of a vacuum
(1.257×10^{-6} Henry/m)
Note: not the same as **permittivity**

Generation of magnetic field

- **In a solid material**

A magnetic field is induced in the material



B = Magnetic Induction (tesla) inside the material

$$B = \mu H$$

Permeability of a solid

Units: Henries per meter (H/m)

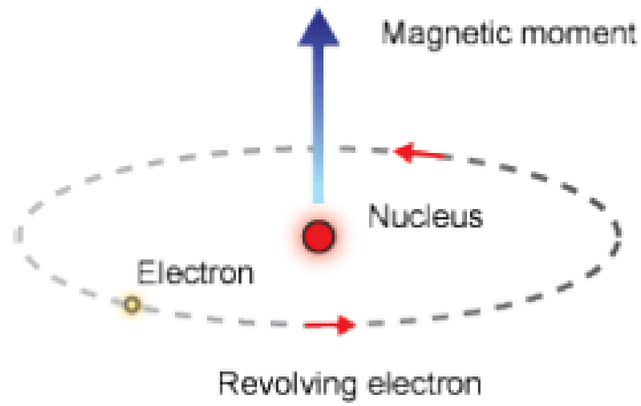
Relative permeability (dimensionless)

Note: not the same as **relative permittivity**

$$\mu_r = \frac{\mu}{\mu_0}$$

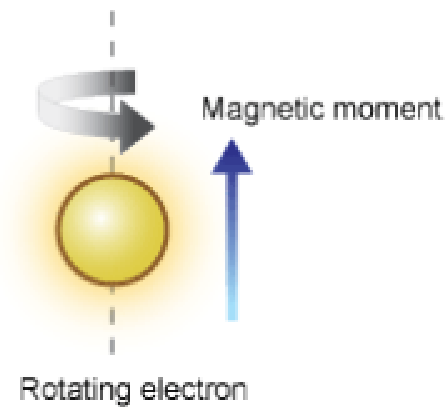
Magnetic moments

- Origin of **magnetic moments**



- Magnetization is created by electrons revolving around a nucleus
  electron orbital motion

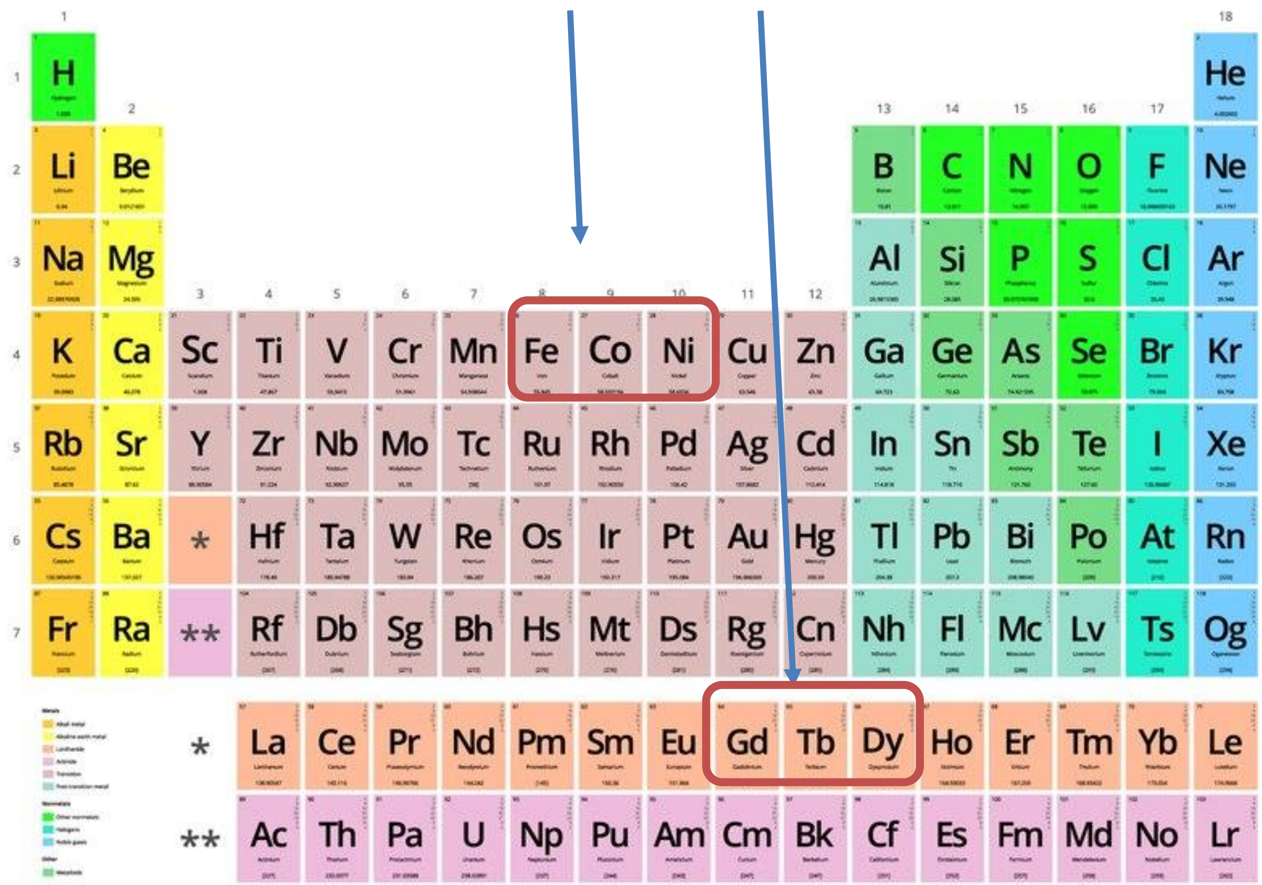
- Magnetization is created by unpaired electrons rotating themselves, m_s
  electron spin (spin up↑and spin down↓)



- The sum of orbital magnetic moment and electron spin magnetic moment from all electrons in the atom gives the property of magnetization

Magnetic elements

Elements with unpaired electrons



Noble gases do not show magnetic moment → no unpaired electrons

Magnetic moments occur in atoms with unpaired electrons and various valences

Generation of magnetic field

- **In a solid material**

Magnetization $M = \chi_m H$

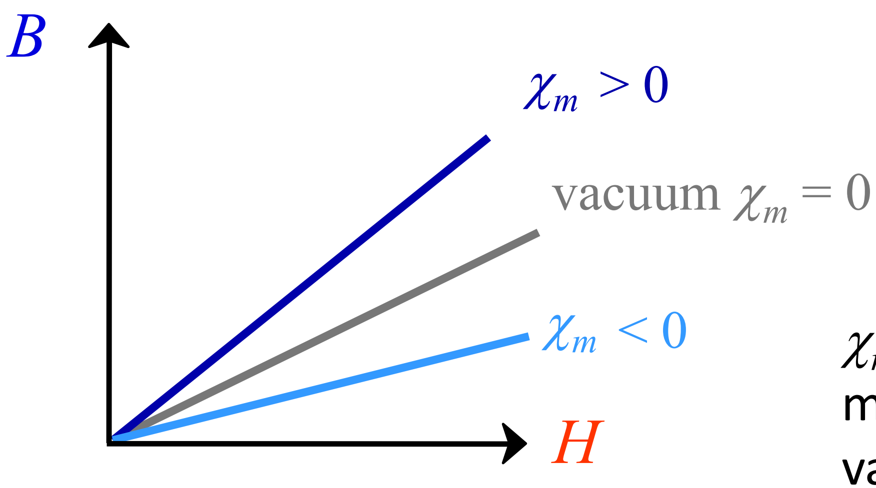
Magnetic susceptibility (dimensionless)

B in terms of H and M $B = \mu_0 H + \mu_0 M$

Combining the above two equations:

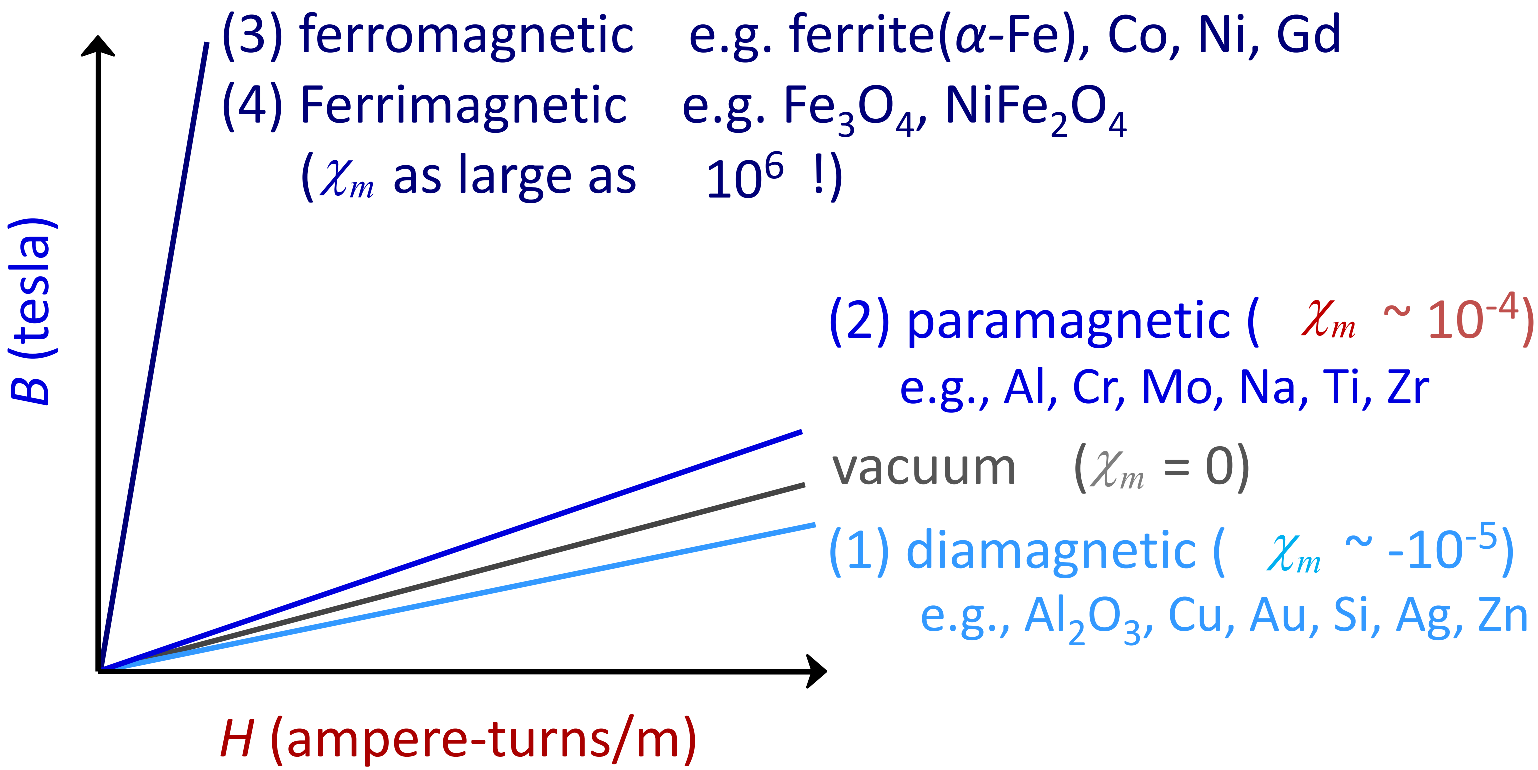
$$B = \mu_0 H + \mu_0 \chi_m H$$
$$= (1 + \chi_m)\mu_0 H$$

permeability of a vacuum: (1.26×10^{-6} Henry/m)



χ_m is a measure of a material's magnetic response relative to a vacuum. $\chi_m = (\mu_r - 1)$

Types of magnetism



Plot adapted from Fig. 20.6, *Callister & Rethwisch 9e*.
Values and materials from Table 20.2 and discussion in Section 20.4.

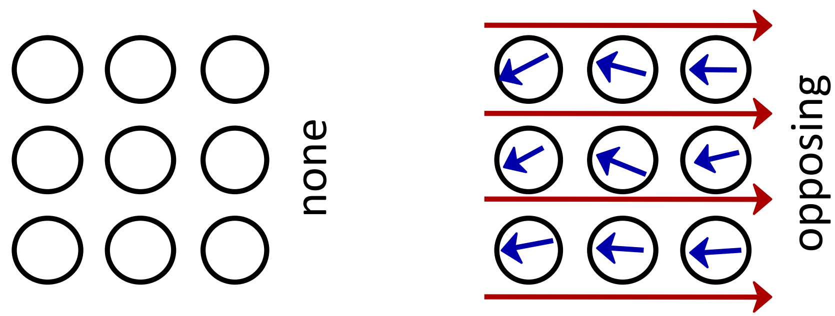
Magnetic responses

Diamagnetic and paramagnetic materials are considered non-magnetic because they only show magnetization under an applied external field

No Applied Magnetic Field (H = 0)

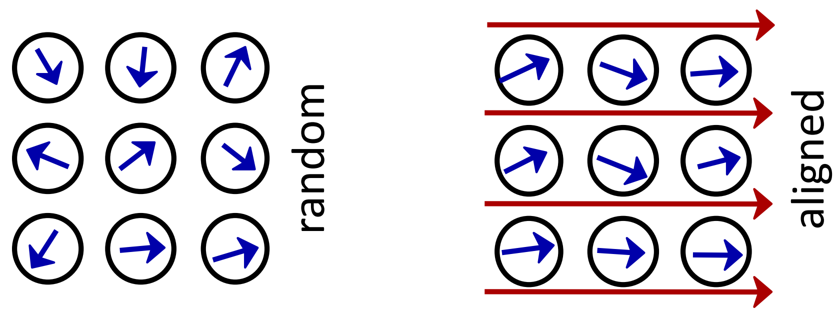
Applied Magnetic Field (H)

(1) diamagnetic



Adapted from Fig. 20.5(a), *Callister & Rethwisch 9e*.

(2) Paramagnetic
Random, permanent



Adapted from Fig. 20.5(b), *Callister & Rethwisch 9e*.

Magnetic responses

- **Interaction between atoms - paramagnetic**

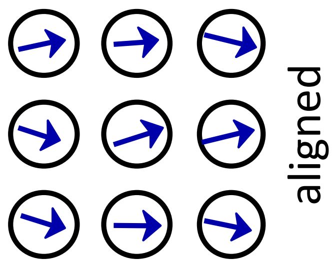
No alignment → no net magnetization

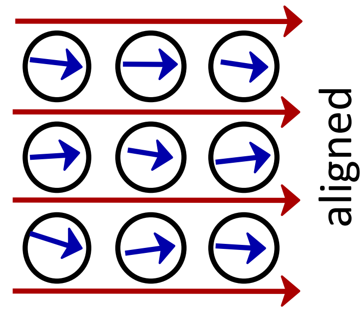
Magnetic responses

No Applied
Magnetic Field (H = 0)

Applied
Magnetic Field (H)

(3) ferromagnetic
(4) ferrimagnetic





Adapted from Fig. 20.7, *Callister & Rethwisch 9e*.

$H << M$ therefore →

$$B \cong \mu_0 M$$

When all magnetic dipoles are aligned with the external field →

$$M_s = n'_B \, \mu_B \, N$$

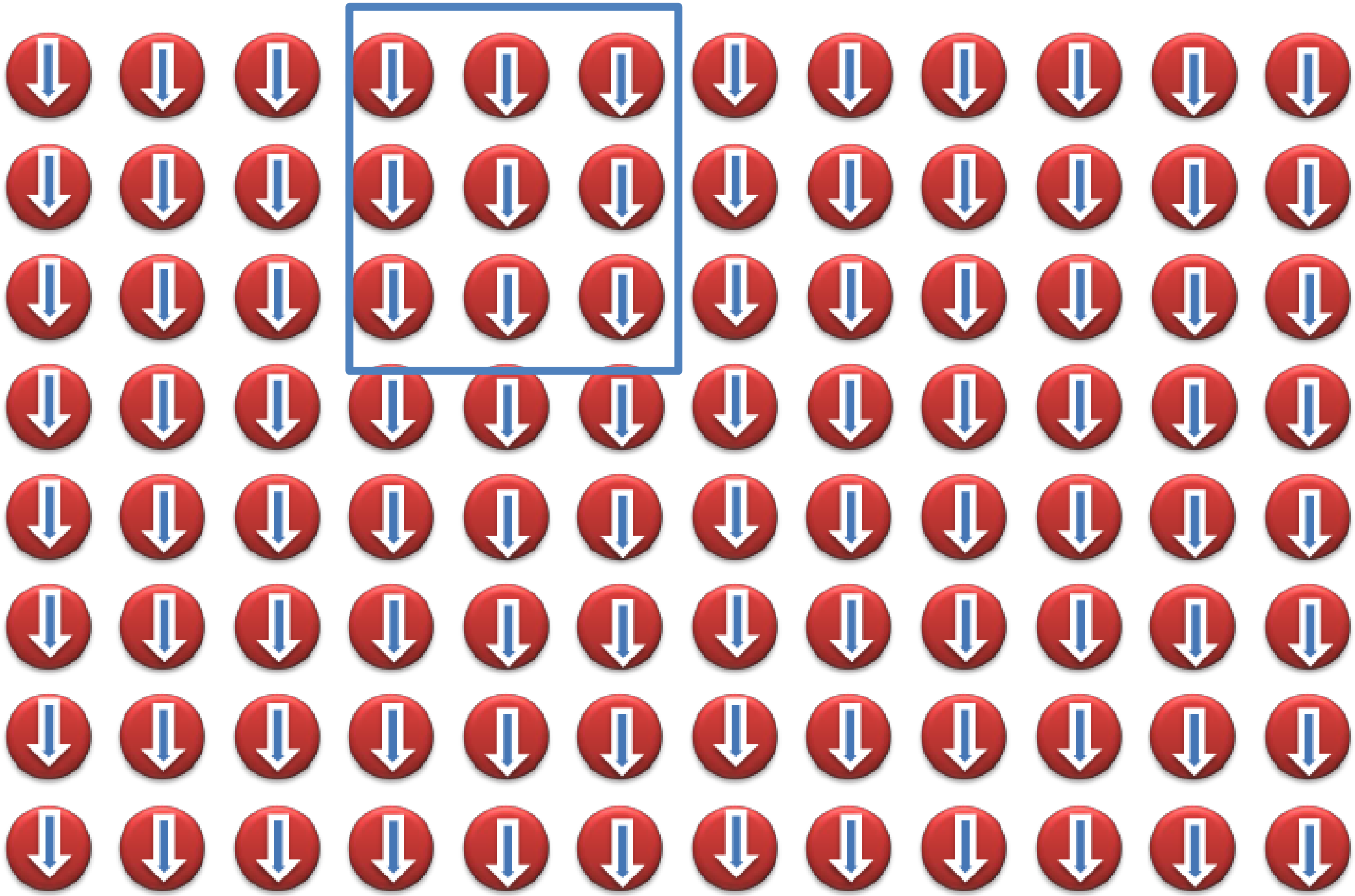
n'_B = net magnetic moment
μ_B = magnitude of Bohr magneton
N = number of atoms

Value of Bohr magnetron μ_B = 9.274×10^{-24} J/T

Magnetic responses

- **Interaction between atoms - ferromagnetic**

Electrons aligned → permanent magnet

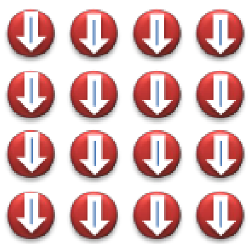


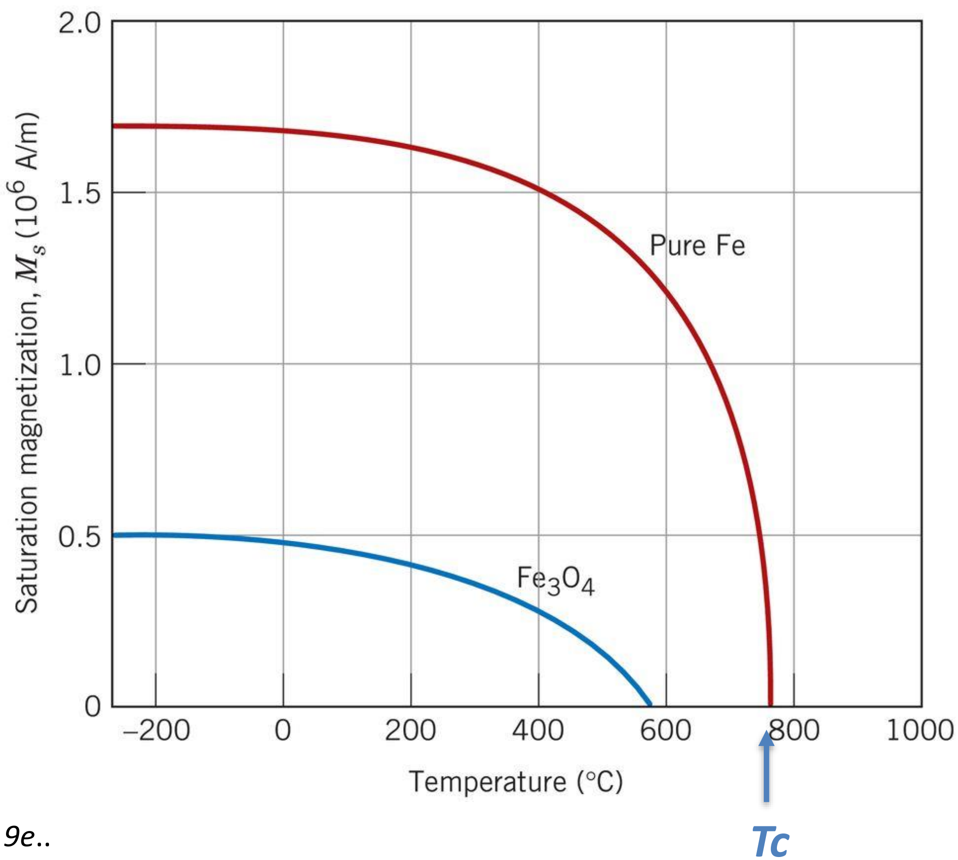
Magnetic responses

- Influence of temperature

At high temperature ferromagnetic materials become paramagnetic above their Curie temperature, *Tc*

(this has a similar meaning to Curie Temperature for ferroelectric materials)





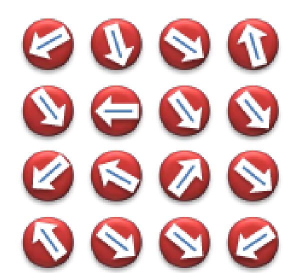
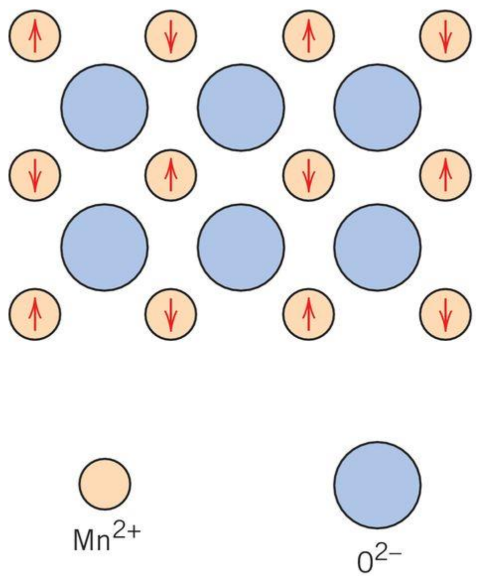
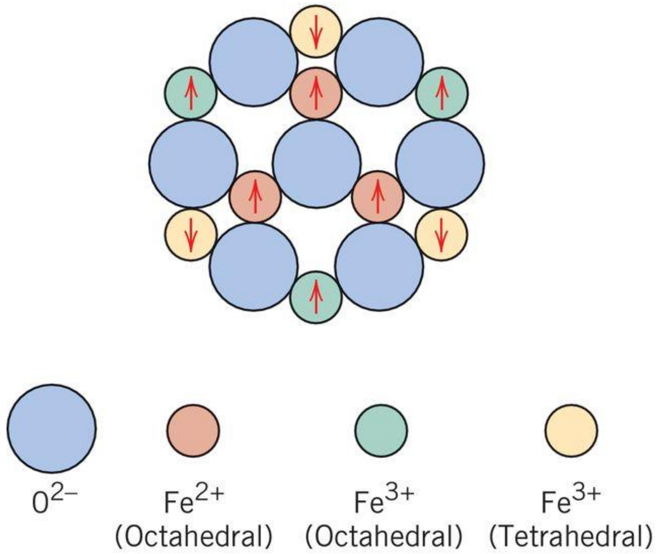


Fig. 21.10, *Callister & Rethwisch 9e*..

Magnetic responses

Antiferromagnetism and **Ferrimagnetism**





Example: MnO (ceramic)
The Mn^{2+} ions have magnetism from electron spin.
Magnetic moments are antiparallel so there is no net magnetism

Example: Fe_3O_4 (magnetite ceramic)
The Fe^{3+} ions in the octahedral and tetrahedral lattice sites have moments that are antiparallel
Fe^{2+} ions in the octahedral sites are ferromagnetic

Fig. 21.8 and 21.9, *Callister & Rethwisch 9e*..

Summary

Magnetic properties:

- Magnetic dipoles and their atomic origin
- Types of magnetic behaviour
  - diamagnetic
  - paramagnetic
  - **ferromagnetic**
  - **ferrimagnetic**
  - antiferromagnetic

Useful magnetic materials